

Design and Implementation of CSI Controller IP

I2C Slave

I²C Slave Module

I²C Slave with Register Interface

Purpose:

- Handle I²C protocol
- Communicate with processor
- Translate I²C transactions into register access

Responsibilities:

- START / STOP detection
- Slave address decoding
- ACK generation
- Read / Write support

CSI TX Register Bank Module

I²C Slave with Register Interface

Purpose:

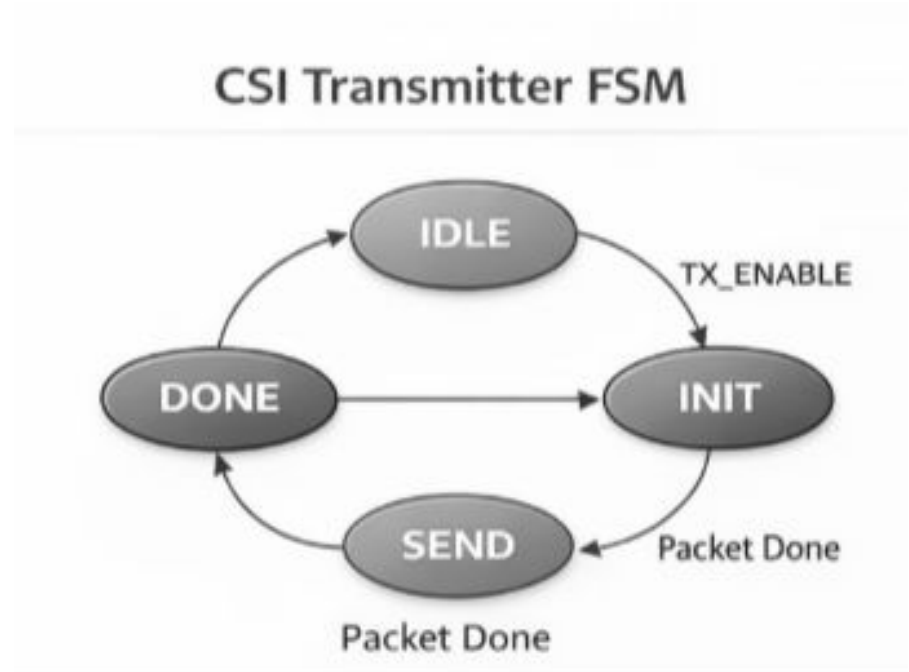
- Store CSI configuration parameters
- Provide control signals to CSI FSM

Address	Name	Access	Bit(s)	Default	Description
0x0000	TX_ENABLE	Write	[0]	0	Enables CSI transmitter when set to 1
0x0001	LANE_COUNT	Write	[1:0]	01	Number of active CSI lanes
0x0002	DATA_TYPE	Write	[5:0]	0x2A	CSI packet data type
0x0100	TX_STATUS	Read	[0]	0	CSI transmitter active status

CSI Transmitter FSM Module

Purpose:

- Control CSI transmission sequence
- Organize initialization and sending phases

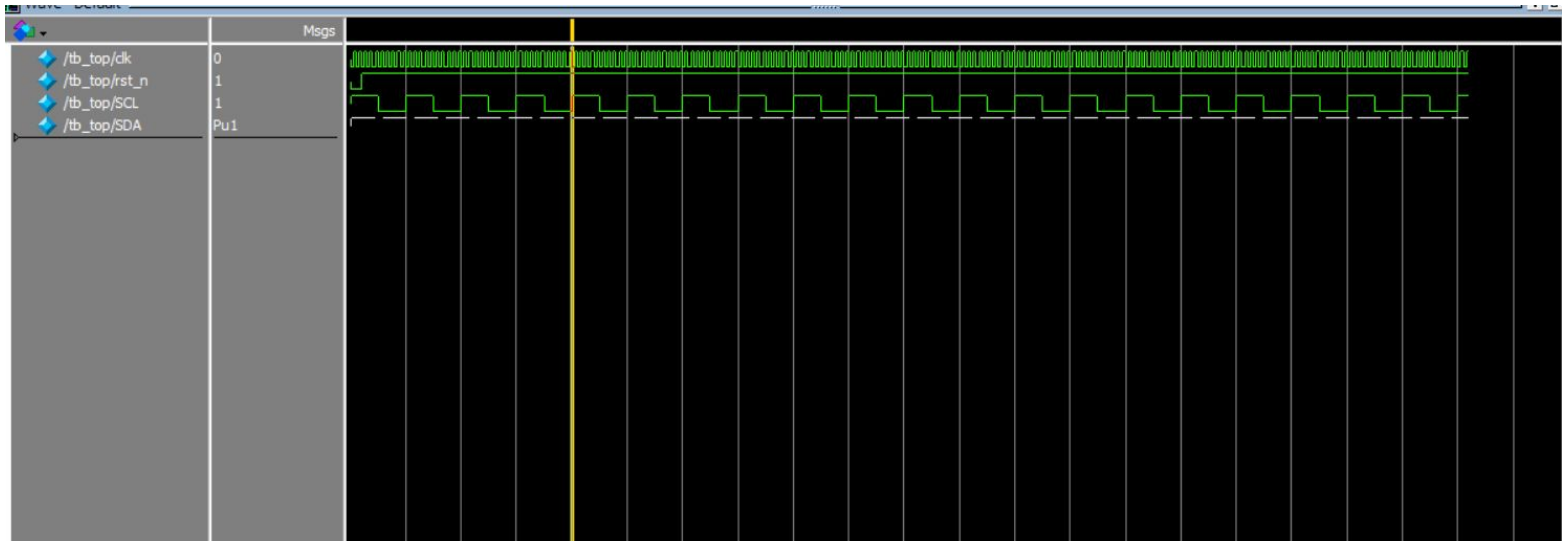


Top Module (Integration Layer) Module

Purpose:

- Connect all modules together
- No protocol or control logic inside

wave



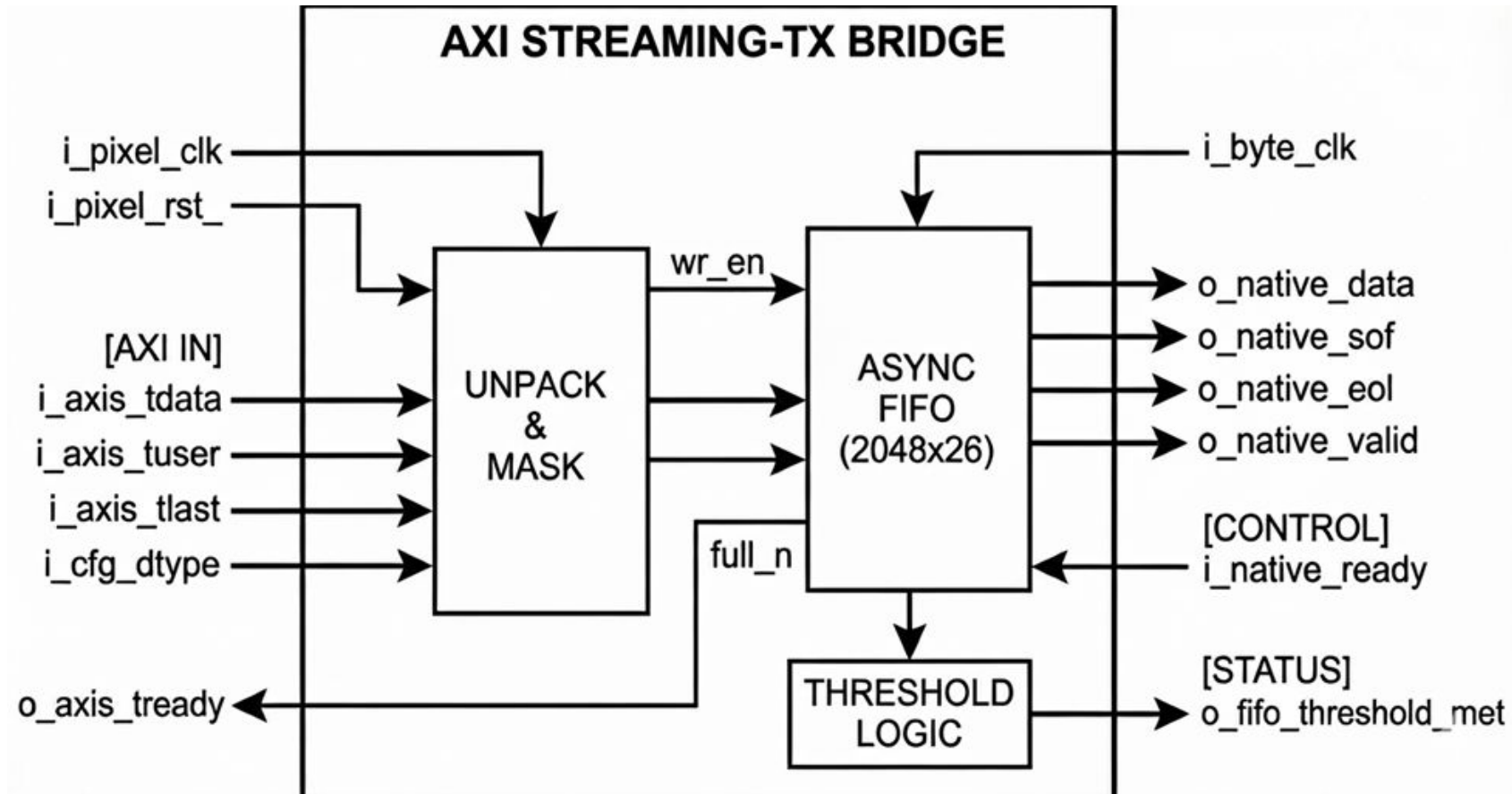
AXI Streaming -TX

4.1.1 Parameters Names

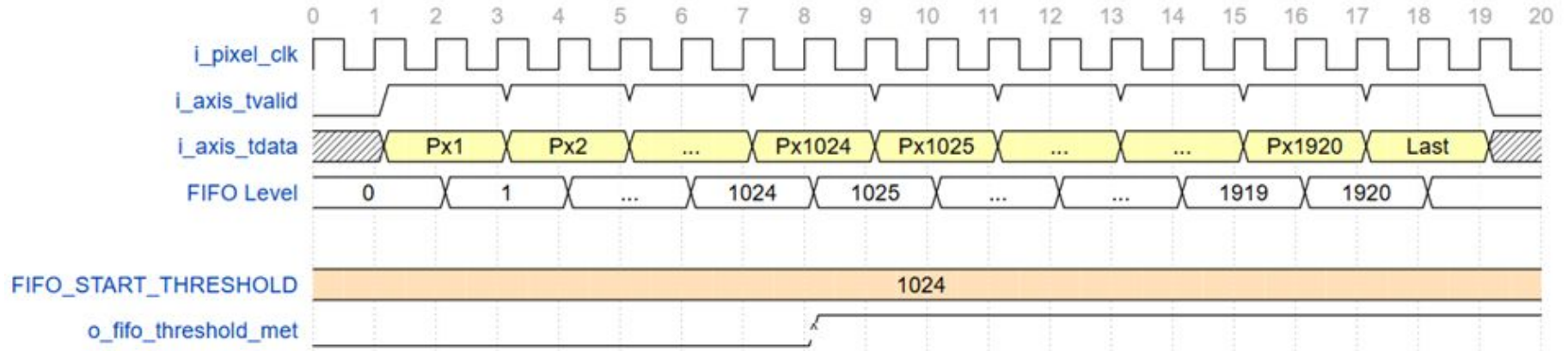
Parameter Name	Default	Description
AXIS_TDATA_WIDTH	32	Width of the AXI input bus. Fixed at 32 to support 24-bit RGB888 inside a word-aligned container.
FIFO_DEPTH	2048	Depth of the internal Asynchronous FIFO. Must be sufficient to buffer incoming pixels during MIPI "Low Power" blanking periods.
FIFO_WIDTH	26	The maximum width of a single valid pixel plus the two sideband signals(tuser and tlast).
MAX_PIXEL_WIDTH	24	The maximum width of a single valid pixel (e.g., 24 for RGB888).
STORE_FULL_LINE	1	Primary Mode. 1: Wait for TLAST (End of Line) before starting. Safest for preventing underflow. 0: Use the Threshold below to start transmission earlier for lower latency.
FIFO_START_THRESHOLD	1300	FIFO Level Trigger. Usage depends on STORE_FULL_LINE:

Port Name Port Width Port Type Description

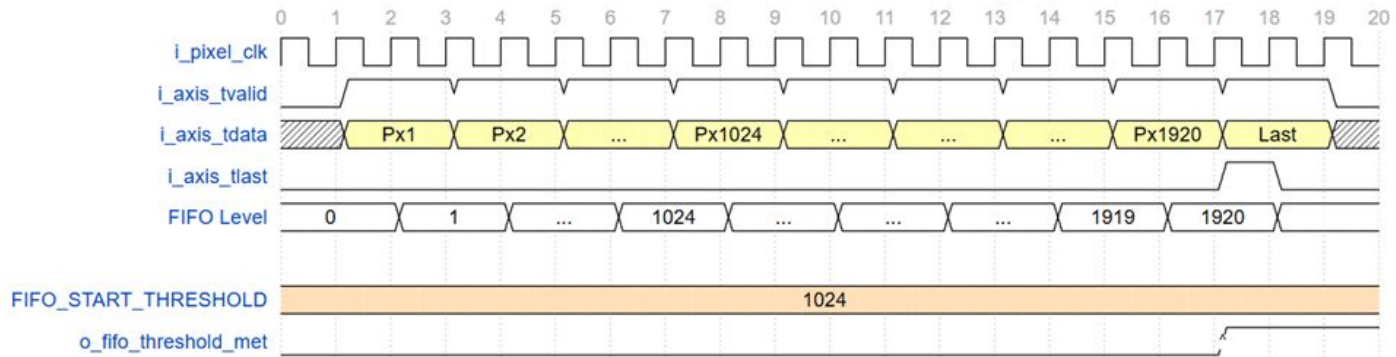
Port Name	Port Width	Port Type	Description
Clocks and Resets			
i_pixel_clk	1	Input	The User/Sensor clock (Write domain).
i_pixel_rst_n	1	Input	Active-low asynchronous reset.
l_byte_clk	1	Input	The MIPI Byte clock (Read domain).
AXI4-Stream Input			
i_axis_tdata	32	Input	Data container holding the pixel + padding.
i_axis_tvalid	1	Input	Valid data indicator.
o_axis_tready	1	Output	Backpressure signal (FIFO Full).
i_axis_tuser	1	Input	Start of Frame (SOF) marker.
i_axis_tlast	1	Input	End of Line (EOL) marker.
Configuration			
i_cfg_data_type	6	Input	MIPI Data Type code (masks valid bits).
Native Output			
o_native_data	24	Output	Clean pixel data to Pixel-to-Byte Converter.
o_native_valid	1	Output	Valid Indicator
i_native_ready	1	Input	Read enable from downstream.
o_native_sof	1	Output	Start of Frame flag.
o_native_eol	1	Output	End of Line flag.
Status Flags			
o_fifo_threshold_met	1	Output	Status signal to the Controller. Goes HIGH only when FIFO fill level > FIFO_START_THRESHOLD (or if line is finished). Indicates safe to start transmission.

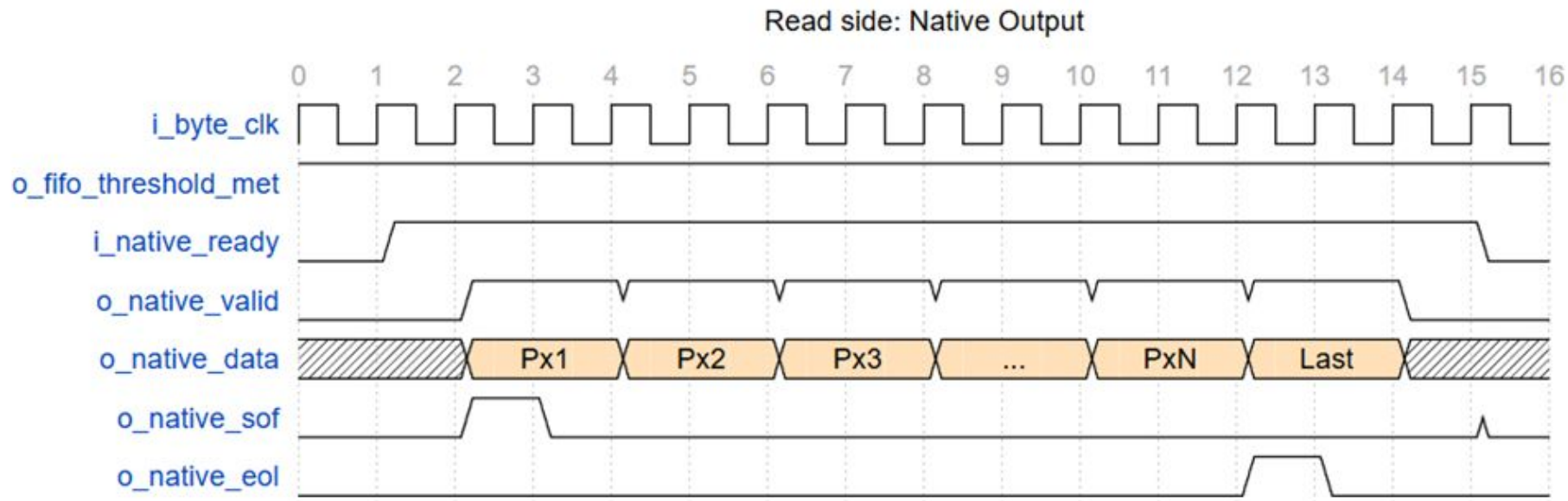


Mode 0: Streaming (Trigger on Threshold)



Mode 1: Store Full Line (Trigger on TLAST)





Pixel to Byte Converter-TX

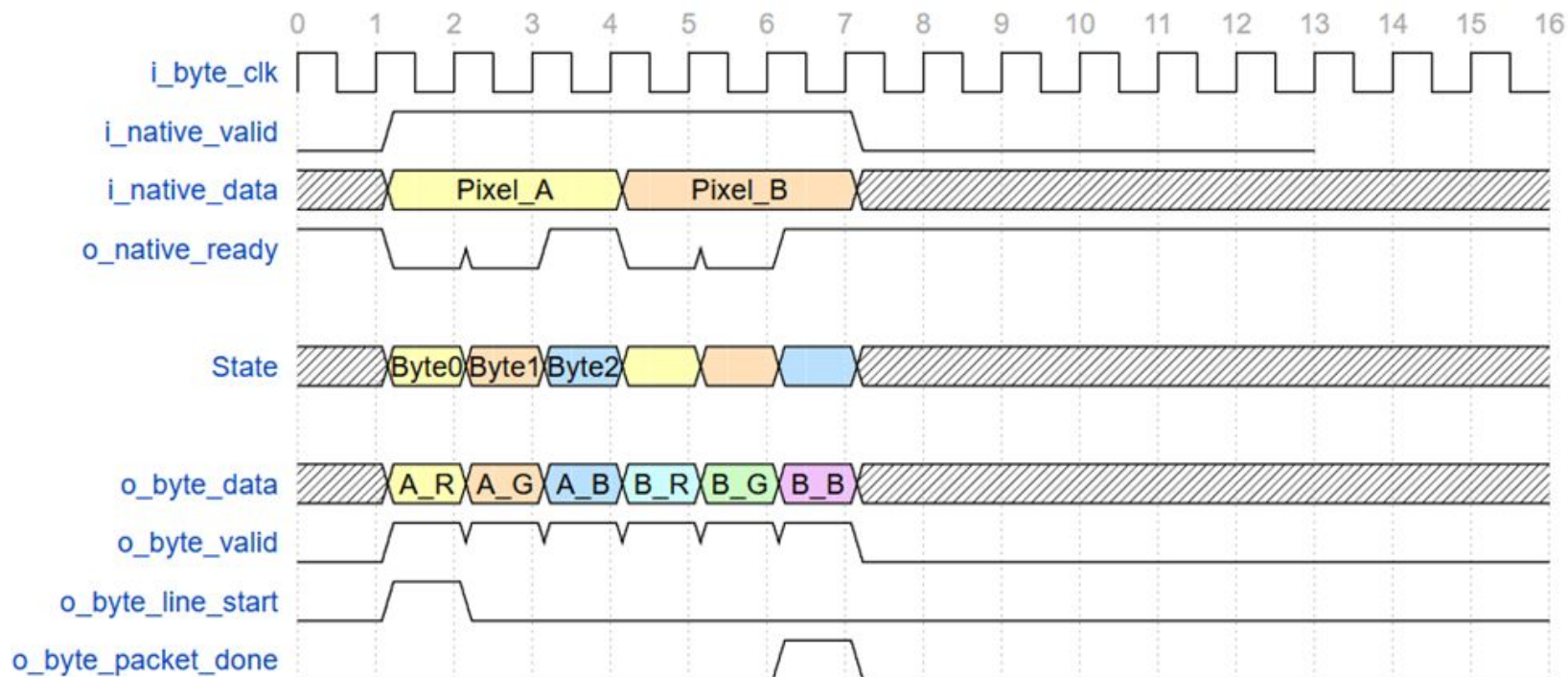
4.1.1 Parameters Names

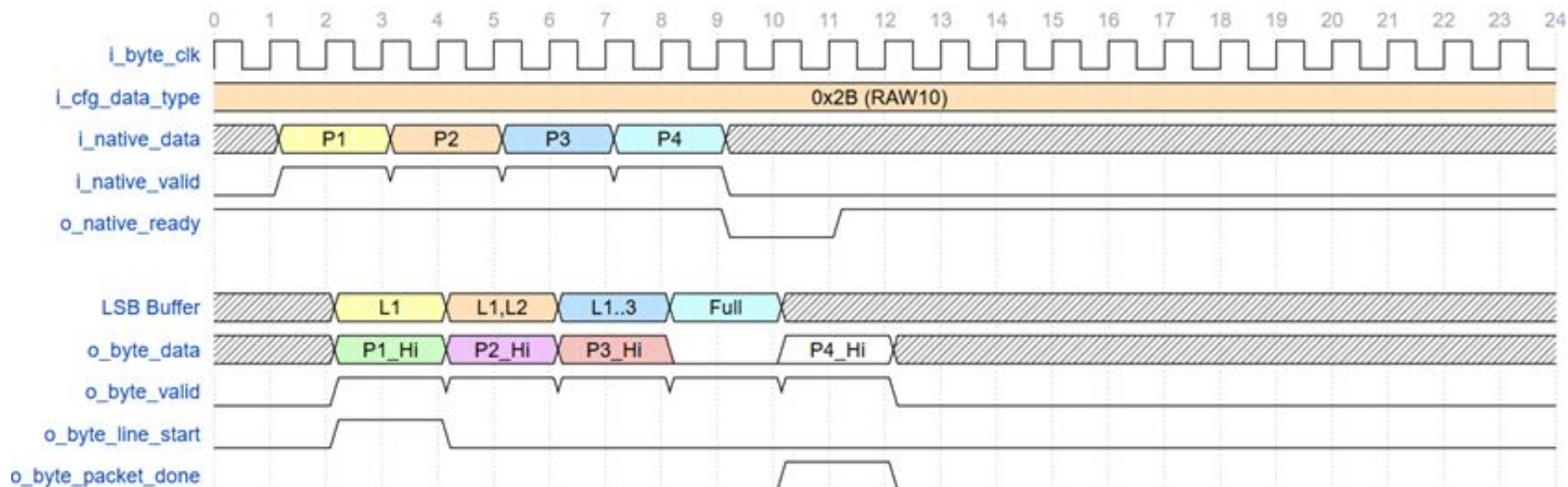
Parameter Name	Default	Description
MAX_PIXEL_WIDTH	24	Maximum input width (e.g., RGB888 needs 24 bits).
OUTPUT_WIDTH	8	Output width to the D-PHY (Standard MIPI byte lane).

4.1.2 Ports Names

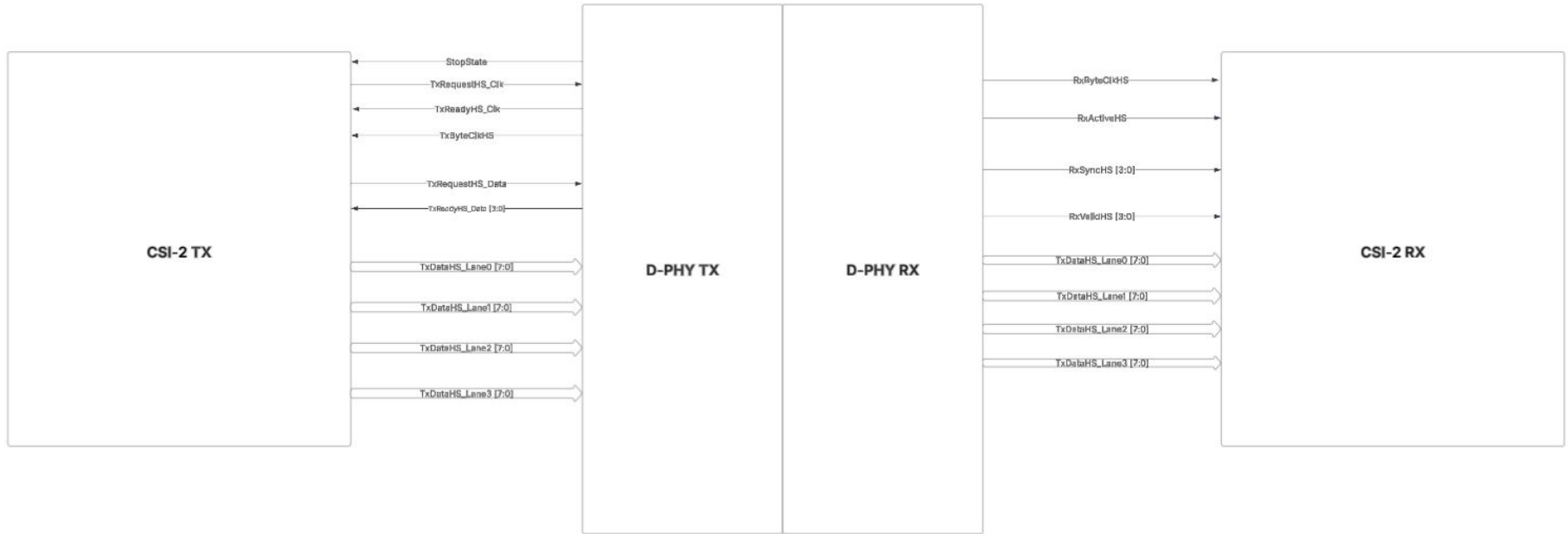
Port Name	Port Width	Port Type	Description
i_cfg_data_type	6	Input	MIPI Data Type (DT) code. Determines the byte count per pixel.
Clocks and Resets			
i_byte_clk	1	Input	The high-speed MIPI Byte clock. All logic in this block runs on this clock.
i_rst_n	1	Input	Active-low asynchronous reset.
Native Input (From AXI Streaming-TX)			
i_native_data	24	Input	Parallel pixel data (RGB).
i_native_valid	1	Input	Valid indicator for the input pixel.
o_native_ready	1	Output	Backpressure Signal. Goes LOW while the converter is busy serializing the current pixel.
i_native_sof	1	Input	Start of Frame marker (aligned with first pixel).
i_native_eol	1	Input	End of Line marker (aligned with last pixel).
Byte Output			
o_byte_data	8	Output	Serialized 8-bit data stream.
o_byte_valid	1	Output	Valid indicator for the output byte.
i_byte_ready	1		Backpressure from D-PHY
o_byte_frame_start	1	Output	FS Trigger: High on the first byte of the First Line.
o_byte_frame_end	1	Output	FE Trigger: High on the last byte of the Last Line (requires logic/counter).
o_byte_line_start	1	Output	PH Trigger: High on the first byte of Every Line.
o_byte_packet_done	1	Output	PF Trigger: High on the last byte of Every Line.

Format	Data Type	Input Group	Output Group	Packing Description
RAW8	0x2A	1 Pixel	1 Byte	P1[7:0] -> Byte0
YUV422 / RGB565	0x1E / 0x22	1 Pixel	2 Bytes	P1[15:8] -> Byte0 P1[7:0] -> Byte1
RGB888	0x24	1 Pixel	3 Bytes	P1[23:16] -> Byte0 P1[15:8] -> Byte1 P1[7:0] -> Byte2
RAW10	0x2B	4 Pixels	5 Bytes	Standard CSI-2 Packing: Bytes 0-3: MSBs (Bits 9:2) of Pixels 1-4. Byte 4: Aggregated LSBs (Bits 1:0) of Pixels 1-4.

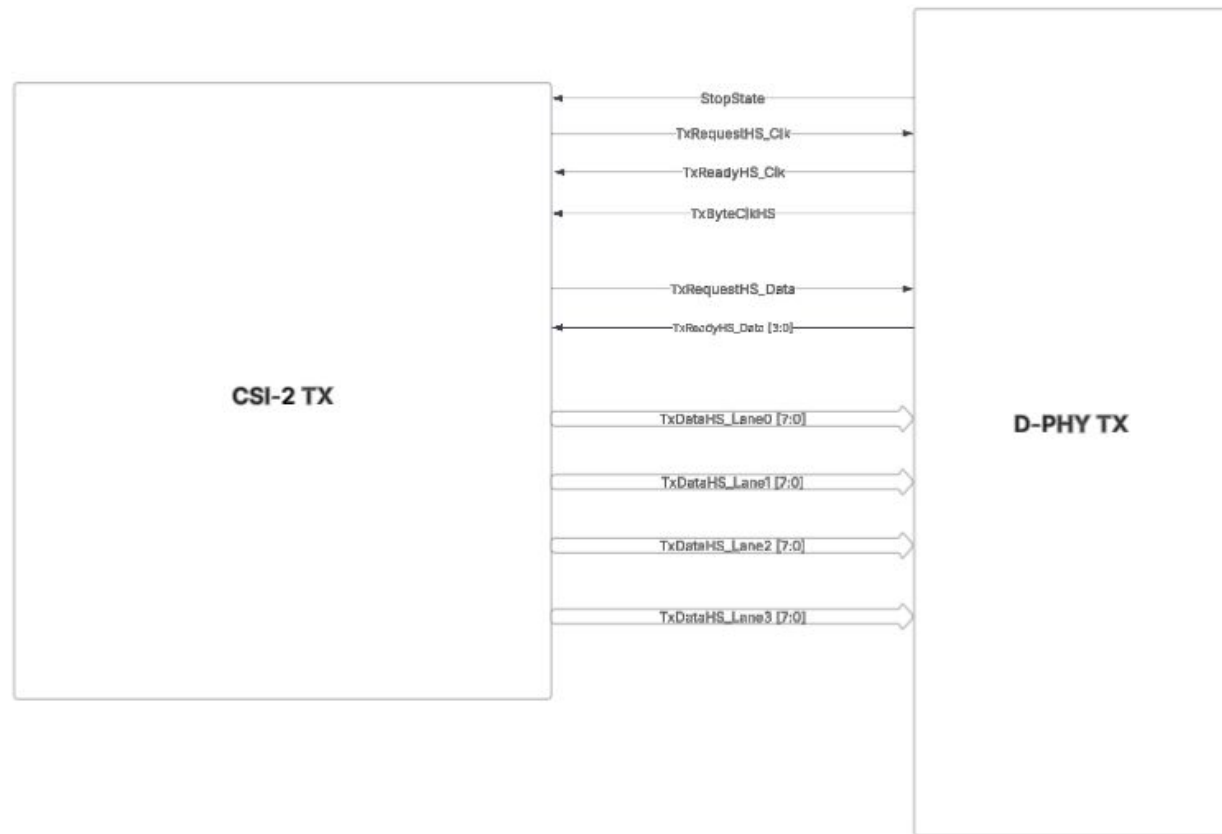




PPI



PPI TX



References

References

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- [2] MIPI Alliance, "Specification for D-PHY Version 2.5," MIPI Alliance, Jul. 5, 2019. [Online].
- [3] Gowin MIPI Byte to Pixel Converter IP
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- [4] Intel Corporation, "*MIPI CSI-2 Intel® FPGA IP User Guide*," Doc. No. 813926, Version 1.1.0, Apr. 26, 2024. Online. Available: <https://www.intel.com/content/www/us/en/docs/programmable/813926.html>
- [5] Gowin MIPI Pixel to Byte Converter IP
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Thank You!